Compounds of Transitional elements

21. Silver nitrate produces a black stain on skin due to
(a) Being a strong reducing agent
(b) Its corrosive action
(c) Formation of complex compound
(d) Its reduction to metallic silver

22. When hypo solution is added to cupric sulphate solution, the blue colour of the latter is discharged, due to formation of
(a) CuS_2O_3 (b) $Na_2S_4O_6$
(c) $NaCuS_2O_3$ (d) Cu_2O

23. Metal oxides which decomposes on heating is
(a) ZnO (b) Al_2O_3
(c) CuO (d) Na_2O
(e) HgO

24. Anhydrous sample of ferric chloride is prepared by heating
(a) $Fe + HCl$
(b) $Fe + Cl_2$
(c) $FeCl_2 + Cl_2$
(d) Hydrated ferric chloride

25. Light green crystals of ferrous sulphate lose water molecule and turn brown on exposure to air. This is due to its oxidation to
(a) Fe_2O_3
(b) $Fe_2O_3.H_2O$
(c) $Fe(OH)SO_4$
(d) $Fe_2O_3 + FeO$

26. In alkaline condition $KMnO_4$, reacts as follows :
$2KMnO_4 + 2KOH \rightarrow 2K_2MnO_4 + H_2O + O$
Therefore its equivalent weight will be
(a) 31.5 (b) 52.7
(c) 72.0 (d) 158.0

27. Equivalent weight of $KMnO_4$ acting as an oxidant in acidic medium is equal to
(a) Molecular weight of $KMnO_4$
(b) $\frac{1}{2}\times$ Molecular weight of $KMnO_4$
(c) $\frac{1}{3}\times$ Molecular weight of $KMnO_4$
(d) $\frac{1}{5}\times$ Molecular weight of $KMnO_4$

28. In which of the following ionic radii of chromium would be smallest
(a) K_2CrO_4 (b) CrO_2
(c) $CrCl_3$ (d) CrF_2

29. $CoO.Al_2O_3$ is called
(a) Cobalt aluminate
(b) Thenard's blue
(c) Both (a) and (b)
(d) None of these

30. $ZnO.CoO$ is called
(a) Cobalt zincate
(b) Rinman's green
(c) Both (a) and (b)
(d) None of these

31. $FeSO_4.(NH_4)_2SO_4.6H_2O$ is called
(a) Mohr's salt

(b) Green salt
(c) Alum
(d) Glauber's salt

32. Molybdenum compounds are used in
(a) Dye industry
(b) For colouring leather
(c) For colouring rubber
(d) All of these

33. When copper turnings and concentrated HCl is heated with copper sulphate the compound formed is
(a) Cupric chloride
(b) Cuprous chloride
(c) Copper sulphate
(d) SO_2

34. The compound of copper which turns green on keeping in air is
(a) Copper sulphate
(b) Copper nitrate
(c) Cupric chloride
(d) Cuprous chloride

35. Cu_2Cl_2 with HCl in presence of oxidising agents gives
(a) $CuCl_2$
(b) H_2CuCl_2
(c) Hydrogen gas
(d) Chlorine gas

36. $K_2Cr_2O_7$ on heating with aqueous $NaOH$ gives
(a) CrO_4^{2-} (b) $Cr(OH)_3$
(c) $Cr_2O_7^{2-}$ (d) $Cr(OH)_2$

37. $KMnO_4$ reacts with oxalic acid according to the equation :
$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$
Here 20 *ml* of 0.1 *M* $KMnO_4$ is equivalent to
(a) $20ml$ of 0.5 *M* $C_2H_2O_4$
(b) $50ml$ of 0.1 *M* $C_2H_2O_4$
(c) $50ml$ of 0.5 *M* $C_2H_2O_4$
(d) $20ml$ of 0.1 *M* $C_2H_2O_4$

38. The equivalent weight of potassium permanganate for acid solution is
(a) 158 (b) 31.6
(c) 52.16 (d) 79

39. Which statement is not correct
(a) Potassium permanganate is a powerful oxidising substance
(b) Potassium permanganate is a weaker oxidising substance than potassium dichromate
(c) Potassium permanganate is a stronger oxidising substance than potassium dichromate
(d) Potassium dichromate oxidises a secondary alcohol into a ketone

40. The formula of corrosive sublimate is
(a) $HgCl_2$ (b) Hg_2Cl_2
(c) Hg_2O (d) Hg